

*System Software*  
*Vs.*  
*Application*  
*Software*

# Computer software

*Computer software, or just software, is a general term primarily used for set of instructions such as computer programs and other kinds of information read and written by computers.*

# Types of Software

- *System Software*
- *Application Software*

# System Software

*System Software includes the Operating System and all the utilities that enable the computer to function.*

**System software** is a term referring to any computer software which manages and controls the hardware so that application software can perform a task.

## Example:

*Operating Systems, Compiler, Loader, Linker, Interpreter.*

# Application Software

*Application Software includes programs that do real work for user.*

## **Example:**

*Payroll systems, Inventory Control, Manage student database, Word Processor, Spreadsheet and Database Management System etc.,*

# *System software Vs. Application Software*

*System software is a Software designed to operate the computer hardware and to provide and maintain a platform for running application software.*

# System Software:

## 1. Operating System

- *Operating System is a software, which makes a computer to actually work.*
- *It is the software that enables all the programs we use.*
- *The OS organizes and controls the hardware.*
- *OS acts as an interface between the application programs and the machine hardware.*
- *Examples: Windows, Linux, Unix, DOS*

# System Software

**2. Compiler:** A compiler is a program that reads a program in one language – the source language and translates into an equivalent program in another language – the target language (Machine understandable Language).

## **Source Language**

'C' language

'Pascal' language

FORTRAN

language C++

language

ADA language

## **Target Languages**

Machine

Understandable language



# System Software

3. **Loader** (contd...): *the part of an operating system*

*that is responsible for loading programs into memory, preparing them for execution and then executing them.*

*The loader is usually a part of the operating system's kernel and usually is loaded at system boot time and stays in memory until the system is rebooted, shut down, or powered off.*

# System Software

4. Linker (contd...): **link editor** is a program that takes one or more objects generated by compilers and assembles them into a single executable program.

*Linkers can take objects from a collection called a library. The objects are program modules containing machine code and information for the linker.*

# System Software

(contd...):

**5. Interpreter:** *An interpreter is a computer program that translates and executes instructions written in a computer programming language line-by-line, unit by unit etc.,*

*An interpreter needs to be able to analyze, or parse, instructions written in the source language.*

Example: *Lisp systems, etc.,*

# Application Software:

## 1. Word Processors:

*Word processing is a tool that helps user in creating, editing, and printing documents. Word processors will normally have the following capabilities built into them:*

- *Spell checking*
- *Standard layouts for normal documents*
- *Have some characters appear in bold print, italics, or underlined*
- *Center lines, make text line up on the left side of the paper, or the right side of the paper*
- *Save the document so it can be used again*
- *print the document.*

*Examples: WordPerfect and Microsoft Word*

# Application Software

**2. Spreadsheets (contd.):** *The spreadsheet packages are designed to use numbers and formulas to do calculations with ease. Examples of spreadsheets include:*

- *Budgets*
- *Payrolls*
- *Grade Calculations*
- *Address Lists*

*The most commonly used spreadsheet programs are Microsoft Excel and Lotus 123.*

# Application

## Software

### **3. Graphic (contd.) Presentations:**

*The presentation programs can make giving presentations and using overheads easier. Other uses include:*

- *Slide Shows*
- *Repeating Computer Presentations on a computer monitor*
- *Using Sound and animation in slide shows*

*The most recognized graphic presentation programs are Microsoft PowerPoint and Harvard*

- *Graphics.*

# Application

## Software

### 4. Database Management System

(Conto...)

- (DBMS) is a software tool that allows multiple users to store, access, and process data into useful information.

- Database programs are designed for these applications:  
types of

- Membership lists
- Student lists
- Grade reports
- Instructor schedules

All of these have to be maintained so you can find what you need quickly and accurately.

- Example: Microsoft Access, dBASE, Oracle.